

# VENKATA SANDEEP KUMAR REDDY

DATA ENGINEER | CLOUD DATA ENGINEERING | STREAMING | ANALYTICS ENGINEERING

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## PROFESSIONAL SUMMARY

Data Engineer with **3 years of professional experience** in SQL, Python, ETL pipelines, data extraction, transformation, validation and data-quality engineering, with hands-on engineering across real-time streaming, cloud data platforms, CDC, lakehouse architecture, modern data warehousing and analytics engineering. Built and validated three end-to-end data platforms using Kafka, Apache Spark, PySpark, PostgreSQL, Apache Iceberg, dimensional modeling, ELT pipelines, data quality, reconciliation, governance, observability and CI/CD, with measurable scale and performance outcomes.

## CORE TECHNICAL SKILLS

**Programming:** Python, SQL, PySpark

**Data Engineering:** ETL Pipelines, ELT Pipelines, Batch Processing, Streaming, CDC, Data Quality, Validation, Reconciliation, Data Modeling, Data Lineage

**Big Data & Streaming:** Apache Kafka, Apache Spark, Spark Streaming

**Data Platforms:** PostgreSQL, Parquet, Apache Iceberg, Data Lakehouse, Data Warehousing

**Cloud:** AWS-oriented Data Engineering, Amazon S3, AWS Glue Data Catalog, Athena, IAM, KMS, CloudWatch, Terraform

**Analytics Engineering:** Dimensional Modeling, Fact & Dimension Tables, SCD, Incremental Models, Data Marts, Semantic Layer, BI-ready Data Products

**Engineering:** Docker, Kubernetes, CI/CD, GitHub Actions, Testing, Observability, Performance & Reliability

**Applications & ML:** FastAPI, React, Streamlit, scikit-learn, Isolation Forest

## PROFESSIONAL EXPERIENCE

**Junior Data Engineer | Mar Symantech Private Limited | September 2023 – September 2026**

- Developed and supported SQL and Python ETL pipelines for data extraction, transformation, validation and downstream reporting/analytics, working with structured datasets and databases.
- Built and maintained batch and distributed data-processing workflows using Apache Spark/PySpark, applying joins, filtering, aggregation, transformation and source-to-target validation across pipeline stages.
- Worked with Kafka-based data ingestion and streaming workflows, supporting the movement and processing of structured data alongside batch-oriented pipelines.
- Used AWS data services, Amazon S3, Airflow, Docker, Kubernetes and CI/CD practices to support data-processing workflows, orchestration, development, testing and deployment activities.
- Implemented data-quality checks, pipeline validation and troubleshooting, investigating failed or inconsistent processing results and collaborating with senior team members to resolve issues and improve workflow reliability.
- Contributed to development, testing, documentation and pipeline support, following team-defined engineering practices and validating data before downstream consumption.

## ENGINEERING PROJECTS

### Enterprise AI-Powered Real-Time EV Fleet Data Platform

- Built an end-to-end real-time EV telemetry data platform using Python, Kafka, Spark, PostgreSQL, FastAPI, Docker and Kubernetes, covering ingestion, distributed processing, persistence and API serving.
- Processed **98,061,214 telemetry records**: 98,060,402 valid canonical records and 812 rejected records; generated 160,774 feature windows and 8,251 AI samples.
- Integrated Isolation Forest with **0.8619 F1** and **1.0000 recall**; validated **20/20 API benchmark requests** at **77.278 ms p95** latency.

### Cloud-Native Lakehouse, CDC & AI-Ready Data Platform

- Engineered a cloud-oriented data lakehouse architecture across PostgreSQL, raw storage, Bronze Parquet, Apache Iceberg, Silver processing and Gold analytical products.
- Implemented and validated PostgreSQL logical CDC using pgoutput with INSERT/UPDATE/DELETE capture, ordering, LSN progression, deduplication, idempotency and checkpoints.
- Validated **15/15 automated engineering tests**, **63/63 data-quality checks** and **60/60 reconciliation checks**; produced 6 Gold, 9 Raw and 9 Bronze datasets.

### RetailIQ — Modern Data Warehouse & Analytics Engineering Platform

- Built a PostgreSQL modern data warehouse from raw/staging layers through dimensional modeling, fact/dimension tables, SCD, incremental models, data marts and semantic views.
- Implemented ELT and analytics-engineering workflows for reusable transformations, analytical business logic and BI-ready products.
- Validated **34/34 data-quality checks** and **9/9 integration tests** with E2E validation passing; built a six-page Streamlit analytics dashboard and implemented lineage, governance, observability and CI/CD.

## EDUCATION

**Bachelor of Technology (B.Tech) — Data Science** | Presidency University | 2019 – 2023

## CERTIFICATIONS & TRAINING

PGP in Data Science and Machine Learning — Intellipaat, 2024 | Linux Certification Training — Intellipaat, 2024 | Data Science Course — Intellipaat, 2023

## PROFESSIONAL LINKS

**LinkedIn:** <https://www.linkedin.com/in/sandeep-reddy-1a9092386>

**GitHub:** <https://github.com/cvsandeep0105-netizen>

**Portfolio:** <https://cvsandeep0105-netizen.github.io/data-engineering-portfolio/>